

hasklib: Stochastic Process Derivations

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Abstract

This document collects the stochastic-calculus results that the stochastic layer of **hasklib** relies on: from the probability-space setup, Brownian motion, Itô SDE, to Geometric Brownian Motion, Ornstein–Uhlenbeck, Arithmetic Brownian Motion, Cox–Ingersoll–Ross, and numerical schemes. Standard measure-theoretic probability is assumed; see Shreve (2004) or Williams (1991) for construction.

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Chapter 1

Foundations: Itô Calculus and the SDE Framework

1.1 Probability space and Brownian motion

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions. The filtration $\{\mathcal{F}_t\}$ represents the information available up to time t .

Definition 1.1 (Brownian motion).

A standard Brownian motion $\{W_t\}_{t \geq 0}$ is a process with

1. $W_0 = 0$ almost surely;
2. Independent increments: for $s < t < u$, $W_u - W_t$ is independent of $W_t - W_s$;
3. Gaussian increments: $W_t - W_s \sim \mathcal{N}(0, t - s)$;
4. Continuous sample paths.

Remark 1.1.

In code, property (3) is what `NormalRng::draw()` supplies: a step of length Δt uses $W_{t+\Delta t} - W_t = \sqrt{\Delta t} Z$ with $Z \sim \mathcal{N}(0, 1)$. This substitution is the single bridge between the continuous theory here and the discrete simulation code.

One might notice (2) and (4) pull in different directions. Independent increments mean each step of the path is a fresh draw, carrying no memory of the past — nothing about $\mathcal{F}_t$ constrains $W_{t+h} - W_t$. Continuity, on the other hand, means the path never jumps: W_{t+h} must stay close to W_t as $h \rightarrow 0$. It is not obvious a single process can do both at once, and the price of doing so turns out to be that the path is extremely rough and nowhere differentiable.

Fixing t and considering the difference quotient over a step $h > 0$. By the Gaussian-increment property, $W_{t+h} - W_t \sim \mathcal{N}(0, h)$, so the difference quotient

$$\frac{W_{t+h} - W_t}{h}$$

is itself Gaussian with mean 0 and variance

$$\text{Var}\left(\frac{W_{t+h} - W_t}{h}\right) = \frac{1}{h^2} \text{Var}(W_{t+h} - W_t) = \frac{h}{h^2} = \frac{1}{h},$$

using that dividing a random variable by a constant h scales its variance by $1/h^2$. Hence the standard deviation of the difference quotient is $1/\sqrt{h}$, which diverges as $h \downarrow 0$: the path moves

by order $\sqrt{h}$ over a step of order h , so no finite slope survives the limit. This informal argument is made rigorous by the Paley–Wiener–Zygmund theorem (?): with probability one, $t \mapsto W_t$ is nowhere differentiable. It also follows that the path has infinite first-order (total) variation on every interval, so integrals $\int f dW_s$ cannot be defined pathwise as Riemann–Stieltjes integrals. Its quadratic variation, however, is finite, and it is precisely this finite second-order variation that the Itô integral and Itô’s lemma are built on.

1.2 Quadratic variation

The property that distinguishes stochastic from ordinary calculus is that Brownian motion accumulates quadratic variation at a nonzero rate.

Proposition 1.1 (Quadratic variation of Brownian motion).

For a standard Brownian motion, the quadratic variation over $[0, T]$ satisfies

$$\langle W \rangle_T = \lim_{\|\Pi\| \rightarrow 0} \sum_k (W_{t_{k+1}} - W_{t_k})^2 = T \quad \text{in } L^2,$$

where Π is a partition of $[0, T]$. Informally this is written

$$(dW_t)^2 = dt.$$

Proof.

Write the partition as $\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$ and abbreviate

$$\Delta t_k = t_{k+1} - t_k, \quad \Delta W_k = W_{t_{k+1}} - W_{t_k}, \quad S_\Pi = \sum_{k=0}^{n-1} (\Delta W_k)^2.$$

“Convergence in L^2 ” means the mean-square error vanishes: $\mathbb{E}[(S_\Pi - T)^2] \rightarrow 0$ as the mesh $\|\Pi\| := \max_k \Delta t_k \rightarrow 0$. We use only two facts about the increments, both directly from the definition of Brownian motion — each $\Delta W_k \sim \mathcal{N}(0, \Delta t_k)$, and increments over disjoint intervals are independent.

Step 1: The mean is exactly T .

Since $\mathbb{E}[\Delta W_k] = 0$, the second moment is the variance:

$$\mathbb{E}[(\Delta W_k)^2] = \text{Var}(\Delta W_k) = \Delta t_k.$$

Summing over k ,

$$\mathbb{E}[S_\Pi] = \sum_k \mathbb{E}[(\Delta W_k)^2] = \sum_k \Delta t_k = T,$$

which holds for every partition, not merely in the limit. Because the mean is exactly T , the mean-square error is the variance:

$$\mathbb{E}[(S_\Pi - T)^2] = \text{Var}(S_\Pi).$$

So the whole statement reduces to showing $\text{Var}(S_\Pi) \rightarrow 0$.

Step 2 Variance of a single term.

Normalise the increment as $\Delta W_k = \sqrt{\Delta t_k} Z$ with $Z \sim \mathcal{N}(0, 1)$, so $(\Delta W_k)^2 = \Delta t_k Z^2$ and

$$\text{Var}((\Delta W_k)^2) = (\Delta t_k)^2 \text{Var}(Z^2).$$

For a standard normal, $\mathbb{E}[Z^2] = 1$ and $\mathbb{E}[Z^4] = 3$, hence $\text{Var}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = 3 - 1 = 2$. (The fourth moment is one Gaussian integration by parts: with the density ϕ satisfying $\phi'(z) = -z\phi(z)$, integrating $\int z^4\phi dz$ by parts gives $\mathbb{E}[Z^4] = 3\mathbb{E}[Z^2] = 3$.) Therefore

$$\text{Var}((\Delta W_k)^2) = 2(\Delta t_k)^2.$$

Step 3 Variance of the sum, then let the mesh shrink.

The increments ΔW_k are independent, so the squared increments $(\Delta W_k)^2$ are independent too, and the variance of a sum of independent variables is the sum of the variances:

$$\text{Var}(S_\Pi) = \sum_k \text{Var}((\Delta W_k)^2) = 2 \sum_k (\Delta t_k)^2.$$

Bound one factor of Δt_k in each term by the mesh:

$$\sum_k (\Delta t_k)^2 \leq \left(\max_k \Delta t_k \right) \sum_k \Delta t_k = \|\Pi\| T.$$

Combining,

$$\mathbb{E}[(S_\Pi - T)^2] = \text{Var}(S_\Pi) \leq 2 \|\Pi\| T \xrightarrow{\|\Pi\| \rightarrow 0} 0,$$

which is exactly the claimed L^2 convergence. □

Remark 1.2. The mean computation ($\mathbb{E}[S_\Pi] = T$) is what makes $(dW_t)^2 = dt$ plausible; the variance computation is what makes it usable. Per step the squared increment $(\Delta W_k)^2$ still fluctuates — its standard deviation $\sqrt{2}\Delta t_k$ is the same order as its mean — but those fluctuations are independent across steps, so summing averages them out at rate $\|\Pi\|$. Inside a sum (equivalently, an integral) the random $(\Delta W_k)^2$ may therefore be replaced by the deterministic Δt_k . That replacement is the entire content of the shorthand $(dW_t)^2 = dt$ used in the next section.

Chapter 2

Geometric Brownian Motion

Chapter 3

The Ornstein–Uhlenbeck Process

Chapter 4

Arithmetic Brownian Motion

Chapter 5

The Cox–Ingersoll–Ross Process

Chapter 6

Numerical Schemes: Euler–Maruyama and Milstein

Bibliography

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